Choose ONE topic, propose it and get approval.

Each topic specifies: what the system does, what data to use for RAG, how to split into two agent systems, and what the MCP server provides. See full project requirements in the separate Requirements document.

#	**Topic**	**Industry**
1	Legal Document Assistant	Legal Tech
2	Real Estate Investment Analyzer	PropTech
3	University Course Advisor	EdTech
4	Medical Information Assistant	HealthTech
5	E-Commerce Customer Support	Retail
6	Restaurant and Food Ordering	FoodTech
7	IT Helpdesk and Infrastructure	Enterprise IT
8	Financial Portfolio Advisor	FinTech
9	Automotive Dealership Assistant	Automotive
10	Recruitment and HR Assistant	HR Tech
11	Smart Agriculture Advisor	AgriTech
12	Travel and Tourism Planner (Lebanon)	Tourism
13	Construction Project Manager	Construction
14	Personal Finance Manager	FinTech
15	Fitness and Training Coach	Health/Wellness
16	Event Planning and Management	Events

17	Cybersecurity Incident Response	InfoSec
18	Supply Chain and Logistics	Logistics
19	Education Content Creator	EdTech

Topic 1: Legal Document Assistant (Lebanese Law)

Industry: Legal Tech

Description

A system that helps lawyers and citizens understand Lebanese legal documents, find relevant laws, and get plain-language explanations.

RAG Data Sources

- Lebanese Labor Law (available in Arabic and English from official government sites)
- Lebanese Commercial Code
- Lebanese Consumer Protection Law
- You can supplement with AI-generated summaries if official documents are hard to parse, but clearly label AI-generated vs. real content
- Source: http://legallaw.ul.edu.lb/ (Lebanese University legal database), official gazette

Agent System A (LangGraph): Legal Research Agent

Supervisor routes to: Document Search Agent (RAG over laws), Explanation Agent (simplifies legal text into plain language), Comparison Agent (compares clauses across different laws).

- **Tools:** search_laws, simplify_text, compare_clauses

Agent System B (Google ADK or other): Case Precedent Lookup

Independent service that searches a separate database of court case summaries. Agent A calls Agent B when user asks about how a law was applied in practice. RAG over case summaries (can be AI-generated realistic case data).

MCP Server: Legal Calendar / Deadline Calculator

- calculate_deadline: given a filing date and applicable law, compute the legal deadline
- get_court_schedule: returns simulated court availability for a given date range

Topic 2: Real Estate Investment Analyzer

Industry: PropTech

Description

A system that helps investors analyze real estate properties, compare investments, and understand market conditions.

RAG Data Sources

- Real estate listings: generate 50-100 properties with location, price, sqm, bedrooms, year built, description, amenities
- Market reports: create 5-10 realistic market analysis reports for different neighborhoods
- Investment guides: general real estate investment principles, ROI calculations, rental yield formulas

Agent System A (LangGraph): Investment Advisor

Supervisor routes to: Property Search Agent (RAG over listings), Financial Analysis Agent (calculates ROI, rental yield, mortgage payments), Market Research Agent (RAG over market reports).

- **Tools:** search_properties, calculator, analyze_investment

Agent System B (Google ADK or other): Comparable Sales Analyzer

Independent service that finds comparable recently sold properties. Takes a property description and returns similar sales with prices. Own RAG pipeline over sales history data.

MCP Server: Mortgage Calculator Service

- calculate_mortgage: principal, rate, term -> monthly payment, total interest
- calculate_roi: purchase price, rental income, expenses -> annual ROI percentage
- estimate_property_tax: property value, location -> estimated annual tax

Topic 3: University Course Advisor

Industry: EdTech

Description

A system that helps students choose courses, plan their schedule, and understand graduation requirements.

RAG Data Sources

- Course catalog: generate 80-100 courses with code, title, description, prerequisites, credits, professor, schedule, capacity
- Graduation requirements: detailed requirements for 3-4 majors (CS, Business, Engineering, Data Science)
- Professor and course reviews: generate realistic review data with ratings and comments

Agent System A (LangGraph): Academic Advisor

Supervisor routes to: Course Search Agent (RAG over catalog), Requirement Checker Agent (validates if a course plan meets graduation requirements), Schedule Builder Agent (checks for time conflicts).

- **Tools:** search_courses, check_prerequisites, check_schedule_conflict

Agent System B (Google ADK or other): Professor and Course Review Analyzer

Independent service with its own RAG over review data. Agent A calls Agent B when student wants to know about course quality. Returns sentiment analysis, common pros/cons, difficulty rating.

MCP Server: Schedule Optimization Service

- find_available_slots: given current schedule, find free time blocks
- suggest_study_plan: given courses, suggest weekly study hours per course based on difficulty
- check_room_availability: room and time -> available or occupied

Topic 4: Medical Information Assistant

Industry: HealthTech

Description

A system that helps users understand medical conditions, medications, and wellness information. NOT for diagnosis. Always includes disclaimer.

RAG Data Sources

- Medical condition summaries: detailed info sheets for 30-40 common conditions (symptoms, causes, treatments, when to see a doctor)
- Medication database: 50-60 medications (name, uses, dosage, side effects, interactions, contraindications)
- Wellness and prevention guides: nutrition, exercise, sleep, mental health
- Use WHO and Mayo Clinic style formatting for structure

Agent System A (LangGraph): Health Information Agent

Supervisor routes to: Condition Lookup Agent (RAG over conditions), Medication Info Agent (RAG over medications), Wellness Advisor Agent (RAG over wellness guides). MUST include strong guardrails: always add medical disclaimer, never diagnose, redirect to doctor for serious symptoms.

- **Tools:** search_conditions, search_medications, check_drug_interactions

Agent System B (Google ADK or other): Drug Interaction Checker

Independent service that checks interactions between multiple medications. Agent A calls Agent B when user asks about combining medications. Own structured database of interactions.

MCP Server: Symptom Triage Service

- assess_urgency: given symptoms, classify as emergency/urgent/routine/self-care
- find_specialist_type: given condition, suggest what type of doctor to see

Topic 5: E-Commerce Customer Support

Industry: Retail

Description

A complete customer support system for an online store with product search, order management, and policy enforcement.

RAG Data Sources

- Product catalog: generate 100+ products across 5-6 categories with name, description, price, specs, stock status, reviews
- Company policies: return policy, shipping policy, warranty, privacy policy, FAQ (15+ pages total)
- Order history: generate 50 sample orders with statuses, tracking, customer info

Agent System A (LangGraph): Support Supervisor

Supervisor routes to: Product Search Agent (RAG over catalog), Policy Agent (RAG over policies), Order Status Agent (tool-based lookup). Must handle: product questions, return requests, order tracking, complaints.

- **Tools:** search_products, search_policies, lookup_order, calculate_refund

Agent System B (Google ADK or other): Product Recommendation Engine

Independent service that takes user preferences and browsing history, returns personalized recommendations. Own RAG over product catalog with a different ranking strategy. Agent A calls Agent B when user says 'What do you recommend?'

MCP Server: Inventory and Shipping Service

- check_stock: product ID -> availability and quantity
- estimate_shipping: destination, weight -> cost and delivery date
- create_return_label: order ID -> return shipping label URL (simulated)

Topic 6: Restaurant and Food Ordering Assistant
Industry: FoodTech

Description

A system that helps users discover restaurants, understand menus, get dietary recommendations, and place orders.

RAG Data Sources

- Restaurant database: generate 30-40 restaurants with name, cuisine, location, rating, price range, hours, full menu with prices, dietary labels
- Dietary guides: detailed info on vegan, vegetarian, gluten-free, halal, keto, low-sodium, allergies
- Food reviews: generate realistic reviews for restaurants and specific dishes

Agent System A (LangGraph): Food Concierge

Supervisor routes to: Restaurant Search Agent (RAG over restaurants), Menu Analyzer Agent (finds dishes matching dietary needs), Review Agent (RAG over reviews).

- **Tools:** search_restaurants, search_menu_items, filter_by_diet, calculate_order_total

Agent System B (Google ADK or other): Nutrition Analyzer

Independent service that analyzes nutritional content of meals. Takes a list of dishes, returns calories, macros, allergens. Own database of nutritional information. Agent A calls Agent B when user asks 'Is this meal healthy?'

MCP Server: Order and Reservation Service

- place_order: restaurant, items -> order confirmation and total
- make_reservation: restaurant, date, time, party size -> confirmation
- check_wait_time: restaurant -> estimated wait in minutes

Topic 7: IT Helpdesk and Infrastructure Agent

Industry: Enterprise IT

Description

A system that helps employees troubleshoot IT issues, manage access requests, and understand IT policies.

RAG Data Sources

- IT knowledge base: 40-50 troubleshooting articles (WiFi, VPN, email, printer, software, password reset, etc.)
- IT policies: security policy, acceptable use, BYOD, data classification, incident response (15+ pages)
- Software catalog: 30+ approved software with name, purpose, installation guide, license type

Agent System A (LangGraph): IT Support Agent

Supervisor routes to: Troubleshooting Agent (RAG over KB articles), Policy Agent (RAG over policies), Software Agent (RAG over software catalog).

- **Tools:** search_kb, search_policies, search_software, create_ticket

Agent System B (Google ADK or other): Access Management Agent

Independent service that handles access requests and permission checks. 'Does user X have access to system Y?' / 'Request access to system Z for user X.' Own database of users, roles, systems, permissions.

MCP Server: Monitoring and Diagnostics Service

- check_service_status: service name -> up/down/degraded with details
- check_user_account: username -> account status, last login, password expiry
- run_diagnostic: issue type -> automated diagnostic steps and results

Topic 8: Financial Portfolio Advisor

Industry: FinTech

Description

A system that helps users understand their investment portfolio, get market insights, and plan financial goals.

RAG Data Sources

- Financial education content: guides on stocks, bonds, ETFs, mutual funds, crypto basics, risk management, diversification, retirement planning (20+ pages)
- Company profiles: generate 40-50 profiles with sector, market cap, P/E ratio, dividend yield, performance, analyst consensus
- Market reports: create 5-10 weekly/monthly market analysis reports

Agent System A (LangGraph): Financial Advisor

Supervisor routes to: Education Agent (RAG over guides), Company Research Agent (RAG over profiles), Market Analysis Agent (RAG over reports). Must include disclaimer: 'Not financial advice.'

- **Tools:** search_education, search_companies, analyze_portfolio, calculator

Agent System B (Google ADK or other): Risk Assessment Agent

Independent service that analyzes portfolio risk. Takes a list of holdings, returns risk metrics (volatility, Sharpe ratio, diversification score). Own calculation logic and benchmark data.

MCP Server: Market Data Service

- get_stock_price: ticker -> simulated current price and daily change
- get_sector_performance: sector -> YTD performance percentage
- calculate_compound_growth: principal, rate, years -> future value

Topic 9: Automotive Dealership Assistant
Industry: Automotive

Description

A system that helps car buyers research vehicles, compare options, understand financing, and book test drives.

RAG Data Sources

- Vehicle inventory: generate 80-100 vehicles with make, model, year, trim, mileage, price, color, features, condition, dealer notes
- Vehicle spec sheets: detailed specs for 20-25 popular models (engine, transmission, fuel economy, safety ratings, warranty)
- Financing guides: loan calculators, lease vs buy, credit score tiers, insurance basics, trade-in process
- Dealership policies: test drive policy, price match guarantee, warranty terms, service packages

Agent System A (LangGraph): Sales Advisor

Supervisor routes to: Inventory Search Agent (RAG over inventory), Comparison Agent (compare 2-3 vehicles side by side), Policy Agent (RAG over dealership policies).

- **Tools:** search_inventory, compare_vehicles, search_policies, calculator

Agent System B (Google ADK or other): Financing Calculator

Independent service handling all financial calculations. Takes vehicle price, down payment, credit score tier, trade-in value. Returns monthly payment options (36/48/60/72 months), lease options, total cost of ownership. Own database of current interest rates by credit tier.

MCP Server: Scheduling and Availability Service

- book_test_drive: vehicle ID, date, time -> confirmation
- check_vehicle_availability: vehicle ID -> in stock / sold / reserved
- get_service_slots: date -> available service appointment times

Topic 10: Recruitment and HR Assistant

Industry: HR Tech

Description

A system that helps HR teams screen candidates, match them to open positions, and answer employee policy questions.

RAG Data Sources

- Job postings: generate 25-30 open positions with title, department, requirements, nice-to-haves, salary range, location, job description
- Company HR policies: 20+ pages covering leave, benefits, code of conduct, remote work, performance review, onboarding, expenses
- Candidate profiles: generate 50-60 resumes with name, skills, experience, education, certifications, desired role, salary expectation

Agent System A (LangGraph): HR Assistant

Supervisor routes to: Job Search Agent (RAG over postings), Policy Agent (RAG over HR docs), Candidate Screener Agent (evaluates candidate fit against job requirements).

- **Tools:** search_jobs, search_policies, screen_candidate, calculate_benefits

Agent System B (Google ADK or other): Skills Matcher

Independent service that takes a job description and returns ranked candidate matches. Uses embedding similarity between requirements and profiles. Own RAG with skill-weighted scoring. Agent A calls B when recruiter asks 'Best candidates for position X?'

MCP Server: Interview Scheduling Service

- find_available_interviewers: role, date -> list of available interviewers
- schedule_interview: candidate, interviewer, datetime -> confirmation
- send_assessment: candidate, assessment_type -> link to online assessment

Topic 11: Smart Agriculture Advisor

Industry: AgriTech

Description

A system that helps farmers make decisions about crop management, pest control, irrigation, and market timing.

RAG Data Sources

- Crop guides: detailed guides for 15-20 crops common in Lebanon/Mediterranean (olives, grapes, citrus, wheat, tomatoes, etc.) with planting calendar, soil requirements, water needs, harvest timing
- Pest and disease database: 30-40 entries with pest name, affected crops, symptoms, treatments (organic and chemical), prevention
- Soil and fertilizer guides: soil types, pH requirements, fertilizer types, composting
- Market information: seasonal price trends, storage guidelines, export requirements

Agent System A (LangGraph): Farm Advisor

Supervisor routes to: Crop Management Agent (RAG over crop guides), Pest Diagnosis Agent (RAG over pest database), Market Agent (RAG over market info).

- **Tools:** search_crops, search_pests, search_market_data, calculate_yield

Agent System B (Google ADK or other): Weather and Irrigation Planner

Independent service that takes crop type, growth stage, and location. Returns irrigation schedule, frost risk assessment, optimal spray timing. Simulated weather data for Lebanese regions (Bekaa, coast, mountains).

MCP Server: Soil Analysis and Resource Service

- analyze_soil: soil sample data -> pH, nutrients, recommendations
- calculate_fertilizer: crop, area, soil type -> fertilizer type and quantity needed
- estimate_water_usage: crop, area, month -> daily water requirement in liters

Topic 12: Travel and Tourism Planner (Lebanon Focus)

Industry: Tourism

Description

A system that helps tourists plan trips to Lebanon, discover attractions, understand logistics, and handle bookings.

RAG Data Sources

- Attraction database: 60-80 attractions across Lebanon (historical sites, restaurants, nature, nightlife, museums, beaches, ski resorts) with description, hours, price, rating, best season, tips
- Practical travel guides: transportation, safety, currency, SIM cards, cultural etiquette, food guide, hiking trails
- Itinerary templates: pre-built plans ('3 days in Beirut', '1 week all-Lebanon', 'Wine country weekend', 'Adventure trip')

Agent System A (LangGraph): Trip Planner

Supervisor routes to: Attraction Search Agent (RAG over attractions), Logistics Agent (RAG over travel guides), Itinerary Builder Agent (builds custom day-by-day plans).

- **Tools:** search_attractions, search_guides, build_itinerary, calculate_budget

Agent System B (Google ADK or other): Restaurant and Dining Recommender

Independent service specialized in Lebanese food and restaurant recommendations. Takes cuisine preference, location, budget, dietary restrictions, group size. Own RAG over detailed restaurant database with menus and reviews. Agent A calls B when building meal portions of itineraries.

MCP Server: Booking and Transportation Service

- book_hotel: location, dates, budget -> options and confirmation
- estimate_taxi: from, to -> price estimate and duration
- get_bus_schedule: route -> departure times and prices

Topic 13: Construction Project Manager

Industry: Construction

Description

A system that helps construction project managers track progress, manage materials, understand building codes, and estimate costs.

RAG Data Sources

- Building codes: summaries of key Lebanese/international codes (structural, electrical, plumbing, fire safety, accessibility, zoning)
- Material database: 100+ materials with name, specs, unit price, supplier, lead time, alternatives, application guidelines
- PM guides: construction phases, inspection checklists, safety protocols, permit requirements, common delays and mitigations

Agent System A (LangGraph): Project Assistant

Supervisor routes to: Code Compliance Agent (RAG over building codes), Material Agent (RAG over materials), Planning Agent (RAG over PM guides).

- **Tools:** search_codes, search_materials, estimate_cost, check_compliance

Agent System B (Google ADK or other): Cost Estimator

Independent service generating detailed cost estimates. Takes task description (e.g., 'build 200sqm concrete slab'), location, quality grade. Returns itemized material costs, labor estimates, timeline. Own database of labor rates and material pricing.

MCP Server: Inventory and Supplier Service

- check_material_stock: material ID -> quantity, warehouse location
- find_supplier: material, quantity -> supplier options with price and delivery time
- track_delivery: order ID -> status and ETA

Topic 14: Personal Finance Manager

Industry: FinTech

Description

A system that helps individuals manage budget, track expenses, plan savings goals, and understand financial products.

RAG Data Sources

- Financial literacy guides: budgeting methods (50/30/20, zero-based), debt management (snowball vs avalanche), savings strategies, emergency fund, retirement basics
- Financial product database: 30-40 products (savings accounts, credit cards, loans, insurance) with provider, rates, fees, pros/cons
- Lebanese banking context: common banks, typical rates, USD/LBP considerations, banking regulations

Agent System A (LangGraph): Financial Coach

Supervisor routes to: Budget Agent (create and analyze budgets), Product Advisor Agent (RAG over financial products), Education Agent (RAG over literacy guides).

- **Tools:** search_guides, search_products, calculator, analyze_budget

Agent System B (Google ADK or other): Expense Analyzer

Independent service taking expense list (category, amount, date). Returns spending breakdown, trends, anomalies, savings opportunities. Compares against benchmarks ('You spend 35% on dining, typical is 15-20%'). Own analysis logic.

MCP Server: Goal Planning Service

- create_savings_plan: goal amount, deadline, current savings -> monthly contribution needed
- calculate_debt_payoff: debts list -> optimal schedule (snowball and avalanche comparison)
- project_retirement: age, savings, monthly contribution -> projected retirement fund

Topic 15: Fitness and Training Coach
Industry: Health / Wellness

Description

A system that creates personalized workout plans, tracks progress, provides exercise guidance, and offers nutrition advice.

RAG Data Sources

- Exercise database: 120+ exercises with muscle group, equipment, difficulty, form description, common mistakes, variations, sets/reps
- Workout programs: 10-15 templates (beginner full body, PPL, upper/lower, home, bodyweight, 5x5, marathon, HIIT, yoga) with weekly schedules
- Nutrition guides: macros, meal timing, pre/post workout, supplements, hydration, meal prep tips, calorie calculations

Agent System A (LangGraph): Training Coach

Supervisor routes to: Exercise Search Agent (RAG over exercises), Program Builder Agent (creates custom programs based on goals/equipment/schedule), Nutrition Agent (RAG over nutrition guides).

- **Tools:** search_exercises, build_workout, search_nutrition, calculator

Agent System B (Google ADK or other): Progress Tracker and Analyzer

Independent service tracking workout history. Takes user workout logs (exercise, weight, reps, date). Returns strength progression, volume analysis, plateau detection, PR tracking. Agent A calls B when user asks 'Am I making progress?'

MCP Server: Body Metrics and Goal Service

- calculate_tdee: age, weight, height, activity level -> total daily energy expenditure
- calculate_macros: TDEE, goal -> daily protein/carbs/fat targets in grams
- estimate_timeline: current weight, goal weight, deficit -> estimated weeks

Topic 16: Event Planning and Management

Industry: Events

Description

A system that helps event planners organize conferences, weddings, corporate events, and parties.

RAG Data Sources

- Venue database: 40-50 venues in Lebanon (hotels, outdoor, restaurants, conference centers) with capacity, price, amenities, location, catering, A/V
- Event planning guides: timeline templates, vendor checklists, budget templates, seating arrangements, runsheets, contingency planning
- Vendor directory: 60+ vendors (caterers, photographers, DJs, florists, decorators, AV companies) with specialty, price range, rating

Agent System A (LangGraph): Event Planner

Supervisor routes to: Venue Search Agent (RAG over venues), Vendor Search Agent (RAG over vendors), Planning Agent (RAG over guides, creates timelines and checklists).

- **Tools:** search_venues, search_vendors, create_timeline, calculate_budget

Agent System B (Google ADK or other): Budget Optimizer

Independent service optimizing budget allocation. Takes total budget, event type, guest count, priority preferences. Returns recommended allocation by category, vendor tier suggestions, cost-saving tips. Own optimization logic with benchmark data.

MCP Server: Booking and Coordination Service

- check_venue_availability: venue ID, date -> available or booked
- send_vendor_inquiry: vendor ID, event details -> inquiry confirmation
- create_guest_list: event ID, guests -> seating assignments and dietary tracking

Topic 17: Cybersecurity Incident Response
Industry: InfoSec

Description
A system that helps security analysts investigate incidents, understand threats, follow response procedures, and document findings.

RAG Data Sources
- Threat intelligence: 40-50 threat profiles mapped to MITRE ATT&CK framework with IOCs, affected systems, remediation steps
- Incident response playbooks: detailed runbooks for phishing, ransomware, data breach, DDoS, insider threat, compromised credentials, web app attacks
- Security policies: incident classification matrix, escalation procedures, communication templates, evidence preservation, regulatory notifications (GDPR breach)

Agent System A (LangGraph): Incident Commander
Supervisor routes to: Threat Intel Agent (RAG over threats, identifies attack type), Playbook Agent (RAG over runbooks, step-by-step guidance), Documentation Agent (generates incident reports).

- **Tools:** search_threats, search_playbooks, classify_incident, generate_report

Agent System B (Google ADK or other): IOC Analyzer
Independent service analyzing indicators of compromise. Takes IP addresses, file hashes, domain names, email addresses. Returns threat score, known associations, recommended blocks. Own database of IOCs and threat intel (simulated).

MCP Server: Security Operations Service
- check_ip_reputation: IP -> risk score, country, known malicious activity
- lookup_hash: file hash -> known malware match or clean
- check_user_activity: username -> recent logins, anomalies, failed attempts

Topic 18: Supply Chain and Logistics Manager

Industry: Logistics

Description

A system that helps logistics managers track shipments, optimize routes, manage inventory, and handle supplier relationships.

RAG Data Sources

- Supplier database: 40-50 suppliers with company, products, location, lead times, reliability score, pricing tiers, MOQ, payment terms
- Logistics guides: shipping methods (sea/air/land), customs procedures, Incoterms (FOB, CIF, DDP), warehousing, demand forecasting, safety stock
- Product/SKU catalog: 80-100 SKUs with name, category, weight, dimensions, storage requirements, stock level, reorder point, supplier

Agent System A (LangGraph): Logistics Coordinator

Supervisor routes to: Inventory Agent (RAG over SKUs, tracks stock), Supplier Agent (RAG over suppliers, finds and compares), Knowledge Agent (RAG over logistics guides).

- **Tools:** search_inventory, search_suppliers, check_stock_levels, calculate_reorder

Agent System B (Google ADK or other): Route and Cost Optimizer

Independent service optimizing shipping routes and costs. Takes origin, destination, weight, dimensions, urgency. Returns route options with cost, transit time, carrier comparison, customs requirements. Own carrier rate database.

MCP Server: Shipment Tracking and Warehouse Service

- track_shipment: tracking number -> current status, location, ETA
- check_warehouse_capacity: warehouse ID -> available space by storage type
- create_purchase_order: supplier ID, items, quantities -> PO confirmation

Topic 19: Education Content Creator for Teachers
Industry: EdTech

Description
A system that helps teachers create lesson plans, generate quizzes, find resources, and adapt content for different learning levels.

RAG Data Sources

- Curriculum standards: detailed curriculum for 3-4 subjects (Math grades 7-12, Science, English, CS) with learning objectives per grade, topic sequences
- Pedagogy guides: Bloom's taxonomy, differentiated instruction, formative assessment, PBL, flipped classroom, inquiry-based learning
- Resource library: 80-100 resources (worksheets, labs, discussion prompts, projects, exercises) with subject, grade, topic, duration, materials needed

Agent System A (LangGraph): Teaching Assistant
Supervisor routes to: Lesson Planner Agent (RAG over curriculum + pedagogy), Resource Finder Agent (RAG over resource library), Quiz Generator Agent (generates questions aligned with learning objectives).

- **Tools:** search_curriculum, search_resources, generate_quiz, create_lesson_plan

Agent System B (Google ADK or other): Content Differentiator
Independent service adapting educational content for different levels. Takes content text, target level (beginner/intermediate/advanced), learning style. Returns adapted version with appropriate vocabulary, complexity, examples. Specialized LLM prompts for educational adaptation.

MCP Server: Assessment and Analytics Service

- grade_quiz: student answers, answer key -> score and feedback per question
- analyze_class_performance: quiz results -> class statistics, struggling topics, review recommendations
- generate_rubric: assignment description, criteria -> detailed grading rubric